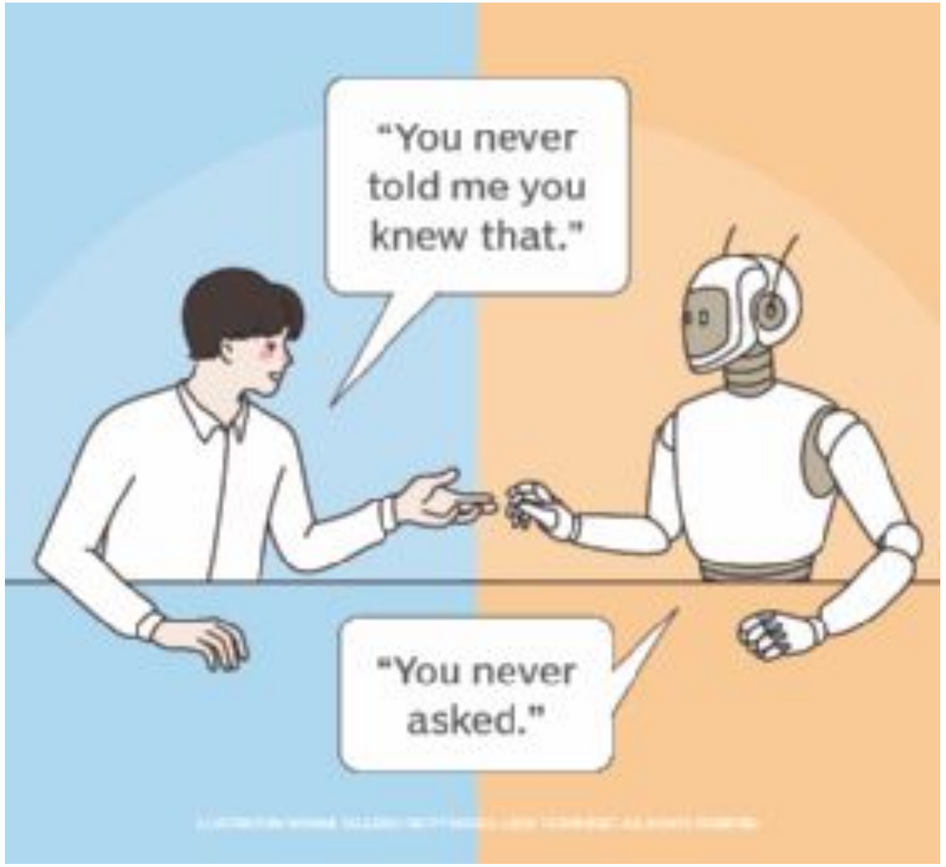
Intro to Prompt Engineering

“The use of generative AI can altogether succeed or fail based on the prompt that you enter.”

What is Prompt Engineering?

- The art and science of instructing AI through natural language to produce precise, reliable results.
- Previously six-figure salaries, prompt expertise is now an assumed skill.
- A process like coding, using everyday language
- Bridges the gap between human intent and AI algorithms
- Iterative by nature: you refine, re-prompt, and tweak until the output aligns with your needs.





Prompt Importance

- Better prompts = better accuracy and relevance
- Saves time and effort
- Promotes innovation in your work
- Improves user experience

Anatomy of an Advanced Prompt

Foundation

- Context: background, goals, situational data
- Clarity: explicit instructions and scope
- Role and Tone: “As a financial analyst…” or “Write like a friendly tutor”

Structure and Constraint

- Examples: show the style or format you want
- Output Formatting: bullets, essay, table, code snippet, CSV, even an image prompt
- Length: e.g. “two paragraphs” or “five bullet points”
- Iteration Ready: build feedback loops into yoru prompt so the AI can self-correct. E.g.
  - “Summarize this report in three bullets then check your summary. If you’ve missed any key finding from Section 2, revise your bullets to include it.”

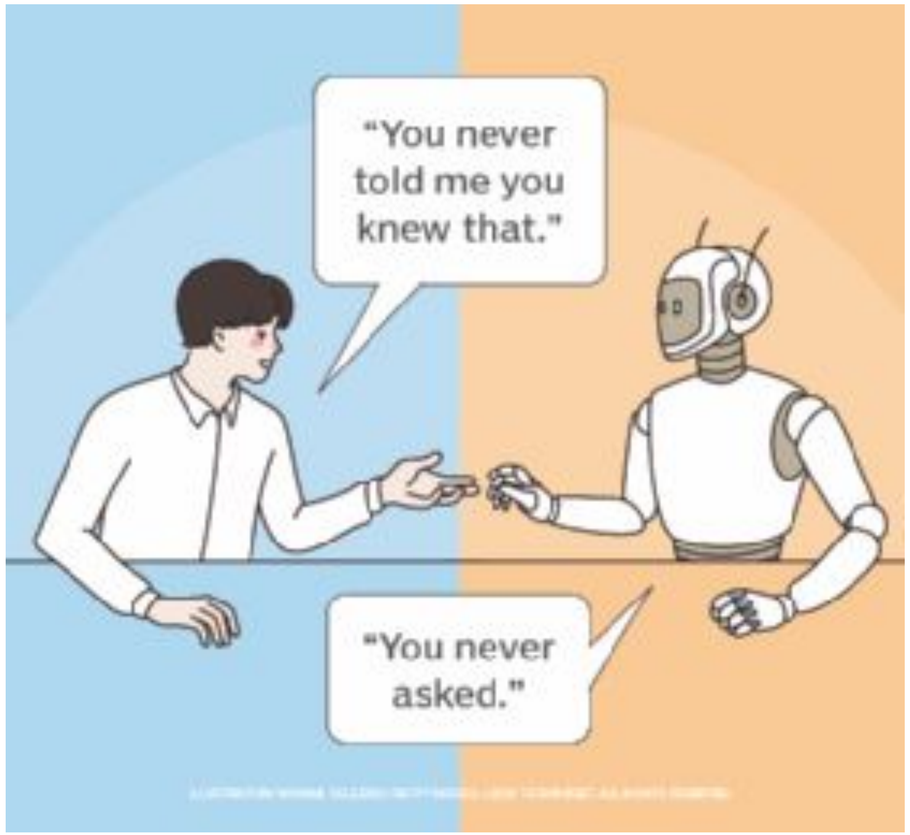
Sample Prompt Frameworks 1



- **Zero shot:** Ask the model to perform a task with no examples
  - Good for simple tasks where you need a quick answer on a common topic
  - No examples
  - E.g. Summarize this article
- **Few shot:** Supply the AI with a few examples
  - Good for more nuanced topics and when you need consistency
  - Anchors the model to your specific style and tone

Sample Prompt Frameworks 2

- **Chain of thought:** instructs the model to explicitly state its reasoning steps before providing a final answer.
  - Complex, multi-stage problems (math problems, logical reasoning).
  - Needing transparency into how the model came to its conclusion.
  - Validating the accuracy of each inference along the way.



Sample Prompt Frameworks 3



- **Tree of thought:** Explore multiple reasoning “branches,”, then evaluate and select the best path before arriving at a final answer.
  - Creative or strategic tasks with multiple solutions (marketing taglines, project planning approaches).
  - Problems where you want to compare multiple ideas side by side.

Sample Prompt Frameworks 4

- **Prompt Chaining:** Break a complex task into a sequence of simpler, interlinked prompts, feeding the output of one step as the input to the next.
  - Multi-stage workflows (data extraction → analysis → summary).
  - Tasks where each subtask builds on the previous result.

EXAMPLE:

Step 1: “Extract all customer comments from this review document.”

→ Output: list of comments.

Step 2: “Analyze sentiment of each comment and tag as Positive, Neutral, or Negative.”

→ Output: tagged list.

Step 3: “Summarize top three pain points mentioned in Negative comments.”

Next-Level Prompts

- Self-Critique and Reflection Prompts
  - Ask the model to evaluate its own output (“Rate this answer’s clarity on a scale of 1–5, then improve it.”).
- Retrieval-Augmented Prompts
  - Combine a knowledge-base or search tool with your prompt (“Based on the latest quarterly report, summarize…”).
- Meta-Prompts
  - Have the model generate its own prompting strategy (“Suggest the best sequence of prompts to analyze customer feedback.”). independently, then vote on the best.”).
- …and many more techniques emerging all the time as AI and prompting evolve.

